Occupied-fibre rigidity and entropy tail nonattainment

New structural results for the B2 midpoint and entropy programs

Research continuation from the supplied handoff

13 October 2026

Abstract

An admissible two-colour remainder in $\mathbb{F}_3^n$ containing no affine plane has at most $\frac{5}{2}\,2^n - 4$ points for $n \geq 3$. The three-dimensional bound of 16 is sharp. The proof establishes a normal form relative to a maximum-dimensional remainder flat: all but at most one nonfull fibre have monochromatic affine coloured complements, and the set of occupied fibres inherits an admissible two-colouring. For the sum–difference entropy ratio, a full-support law on $\mathbb{N}_0$ with $-\log p_n = cn^\beta + o(n^\beta)$ can be stationary at ratio $\rho > 1$ only if $\beta = \log_2(4\rho/(1+\rho))$. Every prescribed positive tail exponent and coefficient nevertheless give a class with the same supremum as the unrestricted problem. In particular the one-sided Gaussian-tail class has the full sharp supremum but no maximizer. No new numerical entropy lower bound or general many-colour codimension improvement is claimed.

1 Scope and baselines

A partition $\mathbb{F}_3^n = A \sqcup B \sqcup R$ is *admissible* if

$$x, y \in A \text{ or } x, y \in B \quad \Longrightarrow \quad (x+y)/2 \notin R.$$

An affine plane has dimension two and nine points. A cap contains no affine line. Empty colour classes are allowed. Density assumptions will be stated separately rather than built into this definition.

The targets are Q4296 (midpoint colourings) and Q4660 (entropy ratio); their stable identifiers are retained in the result ledger. The primary midpoint paper gives an $O(k^2)$ codimension guarantee at density $1/10$ [1]. Continuation I [5] establishes $O(k \log(2k))$ using polynomial Freiman–Ruzsa covering. That general bound is unchanged here. Continuation II [6] establishes the two-colour cap classification and the entropy lower bound

$$1.4051810329 < C_* \leq 2. \tag{1}$$

The upper bound is the established entropic Ruzsa triangle inequality; its sharpness is the question in Section 8 of [2]. The catalog's reference to Conjecture 1.10 concerns a different, sum–product statement. That provenance correction is retained separately without modifying the original records.

The present results have complete written proofs. Exact finite tests are controls on their elementary components, not formal proofs of the all-dimension or infinite-support theorems. Worldwide priority, external peer review, and proof-assistant verification have not been established. Neither complete catalog problem is declared solved.

2 Normal form along a maximum-dimensional remainder flat

We use the following result of Continuation II. A short proof is reproduced in Appendix A to make the present midpoint arguments self-contained.

Lemma 2.1 (Prior cap classification)**.** *A cap that is an admissible two-colour remainder in $\mathbb{F}_3^n$ is empty, a singleton, two points, or a four-point affine parallelogram. In particular, its size is at most four. Here an affine parallelogram is $a + \{0, u, v, u + v\}$ with u, v linearly independent.*

Suppose $R \neq \varnothing$, and let d be the largest dimension of an affine subspace contained in R. Fix the direction $V \leq \mathbb{F}_3^n$ of one such d-dimensional subspace. Let $\pi : \mathbb{F}_3^n \to Q = \mathbb{F}_3^n/V$, and write

$$F_q = \pi^{-1}(q), \qquad R_q = R \cap F_q, \qquad C_q = F_q \setminus R.$$

The full and occupied quotient sets are respectively

$$S = \{q : F_q \subseteq R\}, \qquad D = \{q : R_q \neq \varnothing\}.$$

Thus $S \subseteq D$, and S is nonempty.

Theorem 2.2 (Maximal-flat fibre rigidity)**.** *In the notation above, S has size 1, 2, or 4, and in the last case is an affine parallelogram. All but at most one fibre outside S have a nonempty monochromatic coloured part C_q, and that part is an affine subspace of F_q. Consequently, in every such regular fibre,*

$$|R_q| = 3^d - 3^j \quad \text{for some } 0 \leq j \leq d. \tag{2}$$

If $|S| = 1$ there is no exceptional fibre. If $|S| = 2$ or 4, the only possible exception is the fibre over the unique centre of symmetry of S.

Proof. First, S inherits an admissible colouring: for every $q \notin S$, choose a representative $x_q \in F_q \setminus R$ and assign its colour to q. A same-colour pair whose quotient midpoint lies in S would have its actual midpoint in a fibre entirely contained in R, a contradiction. A line in S would lift to a $(d+1)$-dimensional affine subspace in R. Thus S is a cap and Lemma 2.1 applies.

On $Q \setminus S$, join distinct q, t when $(q + t)/2 \in S$. In that case every point of C_q must have colour different from every point of C_t. Both sets are nonempty. With only two available colours, both are therefore monochromatic.

A vertex $q \notin S$ is isolated precisely when $2s - q \in S$ for all $s \in S$. In characteristic three, $2s - q = 2q - s$, so this says

$$\sigma_q(S) = S, \qquad \sigma_q(s) = 2q - s.$$

Two distinct symmetry centres would give a nonzero translation period of S, by composing the reflections. Its three-point orbits would be affine lines in S, contrary to the cap property. There is at most one isolated vertex. A singleton's centre is its own point, not an exterior vertex; a two-point set or parallelogram has exactly one centre, lying outside it.

Every non-isolated C_q is monochromatic. The midpoint of two of its points is in the same fibre and cannot be in R, so C_q is midpoint-closed. A nonempty midpoint-closed subset of a vector space over $\mathbb{F}_3$ is affine: translate a member to zero, use $\text{midpoint}(0, x) = -x$ to obtain negatives, and then $\text{midpoint}(x, y) = -x - y$ to obtain addition. This proves (2) and the theorem. □

Theorem 2.3 (Occupied-fibre inheritance)**.** *The occupied set D is itself the remainder of an admissible two-colour partition of Q.*

Proof. For $q \notin D$ the entire fibre F_q is coloured. If it is regular, Theorem 2.2 says that it is monochromatic; use that colour for q. In the possible exceptional fibre use any colour that occurs there.

Suppose distinct $q, t \notin D$ receive the same colour i. At most one is exceptional, so one of their entire fibres has colour i and the other has at least one point of colour i. Midpoints of that point with the entire first fibre fill $F_{(q+t)/2}$. Hence the whole midpoint fibre avoids R, and $(q+t)/2 \notin D$. When $q = t$ the conclusion is automatic. This is the required admissibility. □

Remark 2.4. The occupied-fibre conclusion is stronger than inheritance for full fibres. It uses both the two-colour hypothesis and a maximum-dimensional remainder direction. No assertion of occupied-fibre inheritance for arbitrary quotients or arbitrary numbers of colours is made.

3 The sharp plane-free bound in three dimensions

We record two elementary ingredients, distinguishing the classical one from the present deductions.

Lemma 3.1 (Classical affine blocking bound)**.** *A set $C \subseteq \mathbb{F}_3^m$ meeting every affine hyperplane has at least $2m+1$ points. In particular, six points in $\mathbb{F}_3^3$ miss some affine plane.*

Proof. This is the $q = 3$ case of the Jamison–Brouwer–Schrijver blocking theorem [4, 3]; the polynomial proof is included for completeness. Translate a point of C to zero and define, over $\mathbb{F}_3$,

$$P(u) = \prod_{c \in C \setminus \{0\}} (1 - u \cdot c).$$

For $u \neq 0$, the hyperplane $u \cdot x = 1$ contains a point of $C \setminus \{0\}$, so $P(u) = 0$, whereas $P(0) = 1$. Reduce P modulo $u_i^3 - u_i$ in each variable. Reduced polynomials of degree at most two in each variable represent functions on $\mathbb{F}_3^m$ uniquely, as follows by successive one-variable interpolation. The reduced polynomial must be $\prod_i (1 - u_i^2)$, the indicator of zero. It has total degree $2m$; reduction cannot increase total degree. Hence $|C| - 1 \geq 2m$. □

Lemma 3.2 (Small spanning sumsets in $\mathbb{F}_3^3$)**.** *Four affinely independent points have sumset size ten. Five points spanning $\mathbb{F}_3^3$ have sumset size at least eleven. A four-point planar cap has sumset size nine, and an affine line together with an off-line point has sumset size seven.*

Proof. An affine change of coordinates puts an affine basis at $E = \{0, e_1, e_2, e_3\}$. Its ten sums are

$$T = \{0, e_i, 2e_i : 1 \leq i \leq 3\} \cup \{e_i + e_j : i < j\}.$$

For an additional point $v \notin E$, either $2v \notin T$, already adding a new sum, or $v = -e_i$ or $v = -e_i - e_j$. In the first case $v + e_j \notin T$ for $j \neq i$, and in the second $v + e_k \notin T$ for the remaining index k. This proves the five-point assertion. A four-point planar cap is a parallelogram (Appendix A); its sumset fills its plane. Finally, for a line L and $v \notin L$, the three parts $L + L$, $v + L$, and $\{2v\}$ are disjoint, of sizes three, three, and one. □

Theorem 3.3 (Sharp three-dimensional threshold)**.** *In an admissible two-colour partition of $\mathbb{F}_3^3$, a remainder of size at least seventeen contains an affine plane. There is a plane-free remainder of size sixteen.*

Proof. Assume $|R| \geq 17$, and put $C = A \sqcup B$. Then $|C| \leq 10$. Relabel so $a = |A| \geq |B|$. Admissibility gives

$$(A + A)/2 \subseteq C.$$

If $a \leq 3$, then $|C| \leq 6$, and Lemma 3.1 gives a plane in R.

If $a = 4$, then $|C| \leq 8$. The affine-basis and planar-cap possibilities for A are excluded by Lemma 3.2, since their sumsets have sizes ten and nine. Thus A is a line together with an off-line point. Its midpoint set gives seven points of C in a plane H. At most one point of C lies outside H, so at least one of the other two planes parallel to H lies wholly in R.

If $a \geq 5$ and A spans three dimensions, it contains five spanning points. Their sumset has size at least eleven, contradicting $|C| \leq 10$. Otherwise A lies in a plane H and has size at least five. Any two subsets of H with more than half its points intersect, so $(A + A)/2 = H$. Again C has at most one point outside H, yielding a parallel plane in R.

For sharpness take

$$R = (\mathbb{F}_3 \setminus \{0\}) \times (\mathbb{F}_3^2 \setminus \{0\}).$$

Colour points with first coordinate zero by A, and the remaining points with their last two coordinates zero by B. The partition is admissible, and $|R| = 2 \cdot 8 = 16$. If $L \subseteq R$ is affine, each of its two block projections is an affine subspace avoiding zero. Such a projection lies in a level set $\ell = 1$ of a nonzero linear functional: write it as $a + W$ with $a \notin W$ and choose $\ell(W) = 0$, $\ell(a) = 1$. The two pulled-back functionals have disjoint coordinate supports and are independent. Consequently $\dim L \leq 3 - 2 = 1$. A line is obtained by fixing the first coordinate and one coordinate of the second block to one. □

4 An all-dimension plane-free bound

Theorem 4.1. *For every $n \geq 3$, an admissible two-colour remainder $R \subseteq \mathbb{F}_3^n$ containing no affine plane satisfies*

$$\boxed{|R| \leq 20\,2^{n-3} - 4 = \frac{5}{2}\,2^n - 4.} \tag{3}$$

Proof. Let M_n be the maximum size of a plane-free admissible two-colour remainder. Theorem 3.3 gives $M_3 = 16$.

If R has no line, Lemma 2.1 gives $|R| \leq 4$. Otherwise a line in R is a maximum-dimensional remainder flat, since R has no plane. Choose its direction V. In the preceding notation $d = 1$. Every regular nonfull fibre has either zero or two remainder points, by (2), and full fibres have three. At most one occupied fibre can have just one remainder point.

The occupied quotient $D \subseteq \mathbb{F}_3^{n-1}$ is admissible by Theorem 2.3. It is also plane-free. Indeed, if a quotient plane U lay in D, its preimage would be an affine three-space containing nine occupied line fibres. At least eight would contain at least two remainder points and the possible exceptional fibre at least one. The three-space would therefore contain at least seventeen remainder points, contradicting Theorem 3.3.

Every nonfull fibre has at most two remainder points, and there are at most four full fibres. Hence, for $n \geq 4$,

$$|R| \leq 2|D| + |S| \leq 2M_{n-1} + 4.$$

The cap case is smaller as well. Iterating the recurrence from $M_3 = 16$ yields $M_n + 4 \leq 2^{n-3}(16+4)$, proving (3). □

Corollary 4.2 (Explicit density consequence)**.** *An admissible two-colour remainder with $|R| \geq 3^n/10$ contains an affine plane whenever $n \geq 8$. More generally a plane is forced when*

$$|R|/3^n > \frac{5}{2}\left(\frac{2}{3}\right)^n - \frac{4}{3^n}.$$

Proof. At $n = 8$, the upper bound in (3) is 636, below $3^8/10 = 656.1$. The bound divided by 3^n decreases for $n \geq 3$: its successive ratio is $2/3 + 4/(3B_n) < 1$, where $B_n = 20\,2^{n-3} - 4 \geq 16$. □

Remark 4.3. The theorem gives a quantitative extension from line-free to plane-free remainders. It neither determines M_n for $n \geq 4$ nor proves that eight is the smallest dimension forcing a plane at density $1/10$. Its proof is independent of polynomial Freiman–Ruzsa theory.

5 Entropy first variations and tail exponent rigidity

All entropies in this note use natural logarithms; the ratio does not depend on the logarithm base. For independent identically distributed X, X', write

$$\Delta_+ = H(X + X') - H(X), \quad \Delta_- = H(X - X') - H(X), \quad \rho = \Delta_-/\Delta_+.$$

For every nondegenerate discrete real law of finite entropy, $\Delta_+ > 0$. Indeed, $\Delta_+ = I(X; X+X') \geq 0$. Equality would imply that the conditional translated laws of $X + X'$ given any two distinct support points of X coincide, making the law of X' invariant under a nonzero translation. The translates of any positive atom would then give infinite total mass.

Throughout this section the support is *all* of $\mathbb{N}_0 = \{0, 1, 2, \ldots\}$. Put $p_n = \mathbb{P}(X = n) > 0$, and assume

$$-\log p_n = cn^\beta + o(n^\beta), \qquad c > 0, \quad \beta > 0. \tag{4}$$

This assumption implies finiteness of all polynomial moments and of entropy. Define the sum and difference laws by

$$q_s = \sum_{j=0}^{s} p_j p_{s-j} \quad (s \geq 0), \qquad r_m = \sum_{j \geq 0} p_j p_{j+|m|} \quad (m \in \mathbb{Z}).$$

All these probabilities are positive; $r_{-m} = r_m$.

Lemma 5.1 (Coordinate first variation)**.** *For $n \geq 1$, let $p^{(t)} = p + t(e_n - e_0)$, with $|t| < \min(p_0, p_n)$ sufficiently small, where e_j denotes unit mass at j. Define*

$$G_n = -2 \sum_{j \geq 0} p_j \log r_{n-j} + 2\rho \sum_{j \geq 0} p_j \log q_{n+j} - (\rho - 1) \log p_n. \tag{5}$$

Then

$$\frac{d}{dt} \rho(p^{(t)}) \Big|_{t=0} = \frac{G_n - G_0}{\Delta_+(p)}. \tag{6}$$

Proof. Holding $\rho = \rho(p)$ fixed while differentiating gives

$$\Delta_+(p)\, \rho'(0) = \frac{d}{dt}\Big|_{t=0} \left[H(r^{(t)}) - \rho H(q^{(t)}) + (\rho - 1) H(p^{(t)}) \right].$$

The contribution of varying the nth coordinate to the difference entropy is $-2 \sum_j p_j (\log r_{n-j} + 1)$, since the two placements of the varied coordinate give r_{n-j} and r_{j-n}, which agree. For the sum entropy it is $-2 \sum_j p_j (\log q_{n+j} + 1)$. The input term contributes $-(\rho - 1)(\log p_n + 1)$. The coordinate-independent constants cancel between n and 0, leaving (6).

These infinite differentiations are justified as follows. For small enough $|t|$, $p^{(t)}$ is coordinatewise between $p/2$ and $3p/2$; the output laws are correspondingly comparable to q, r. From (4) and the endpoint terms $p_0 p_s$, their negative logarithms are bounded by $K(1 + |s|^\beta)$, uniformly for these t. First derivatives of the output laws are finite linear combinations of translates or reflected translates of p, plus finitely supported terms. They are summable against these polynomial bounds, since all polynomial moments of p are finite. Dominated differentiation therefore applies to all three entropy sums. □

Lemma 5.2 (Convolution tails)**.** *Under (4), set $\kappa_\beta = \min\{1, 2^{1-\beta}\}$. Then*

$$\frac{-\log q_n}{cn^\beta} \longrightarrow \kappa_\beta, \qquad \frac{-\log r_n}{cn^\beta} \longrightarrow 1, \tag{7}$$

$$\frac{-\sum_{j \geq 0} p_j \log q_{n+j}}{cn^\beta} \longrightarrow \kappa_\beta, \qquad \frac{-\sum_{j \geq 0} p_j \log r_{n-j}}{cn^\beta} \longrightarrow 1. \tag{8}$$

Proof. For each $0 < \varepsilon < c$, positive constants $a_\varepsilon, b_\varepsilon$ exist such that for every $j \geq 0$,

$$a_\varepsilon e^{-(c+\varepsilon)j^\beta} \leq p_j \leq b_\varepsilon e^{-(c-\varepsilon)j^\beta}.$$

The minimum of $j^\beta + (n-j)^\beta$ over integer $0 \leq j \leq n$ is asymptotic to $\kappa_\beta n^\beta$. For $\beta \leq 1$ it occurs at an endpoint; for $\beta > 1$ it occurs at one of the two central integers. Using one minimizing summand for a lower bound on q_n and all $n+1$ summands for an upper bound gives the first limit in (7), since $\log(n+1) = o(n^\beta)$ and ε is arbitrary.

For the difference law,

$$p_0 p_n \leq r_n = \sum_{j \geq 0} p_j p_{n+j} \leq b_\varepsilon e^{-(c-\varepsilon)n^\beta} \sum_j p_j.$$

This proves its claimed tail limit. Also $q_s, r_s \geq p_0 p_s$ for $s \geq 0$, so their negative logarithms are bounded by $K(1+s^\beta)$. The integrands in (8), after division by n^β for $n \geq 1$, are bounded by $K'(1+j^\beta)$, using $|n-j| \leq n+j$ and $(n+j)^\beta \leq K_\beta(n^\beta + j^\beta)$. This is integrable under p. For each fixed j the required limit follows from (7). Dominated convergence gives (8). □

Theorem 5.3 (Tail exponent rigidity and explicit ascent)**.** *For a law satisfying* (4)*, its first-variation coefficients obey*

$$\boxed{\frac{G_n}{cn^\beta} \longrightarrow 1 + \rho - 2\rho \min\{1, 2^{1-\beta}\}.} \tag{9}$$

If this limit is positive, moving sufficiently small positive mass from zero to any sufficiently distant atom increases ρ. If it is negative, moving sufficiently small positive mass from any sufficiently distant atom to zero increases ρ.

In particular, if $\rho > 1$ and the law is stationary under all finite coordinate mass transfers—as any local maximum under these transfers must be—then necessarily

$$\boxed{\beta = \log_2\left(\frac{4\rho}{1+\rho}\right) > 1.} \tag{10}$$

Proof. Substitute (8) and (4) into (5). The three coefficients are 2, $-2\rho\kappa_\beta$, and $\rho - 1$; their sum is the limit in (9). Because G_0 is fixed and finite, a nonzero limit determines the eventual sign of $G_n - G_0$. Equation (6) gives the two ascent directions. For each chosen n the transfer can be small enough to preserve positive mass at both atoms, and it does not change the tail asymptotic.

At a stationary law, $G_n = G_0$ for every n. Therefore the limit must vanish. If $\beta \leq 1$ it equals $1 - \rho < 0$, which is impossible. For $\beta > 1$, solving $1 + \rho = 2\rho\, 2^{1-\beta}$ gives (10). □

Corollary 5.4 (Gaussian tails are never stationary)**.** *A full-support law on $\mathbb{N}_0$ satisfying $-\log p_n \sim cn^2$ has a strict ratio-increasing finite-coordinate perturbation whenever $\rho > 1$. Its coefficient in* (9) *is exactly one.*

Proof. For $\beta = 2$, $\kappa_\beta = 1/2$, so $1 + \rho - 2\rho\kappa_\beta = 1$. □

Remark 5.5. The result concerns *one-sided* full support and the specified logarithmic tail law. It makes no claim about two-sided Gaussian laws, finite supports, arbitrary sparse supports, or the existence of an extremizer having a power-law logarithmic tail. In particular, the necessary exponent is not a prediction that a global optimizer exists.

6 The full sharp constant in prescribed tail classes

Let C_* be the supremum of $\rho(P)$ over all nondegenerate discrete real-valued laws with finite entropy. For fixed $\beta, c > 0$, let

$$\mathcal{T}_{\beta,c} = \{P : p_n > 0 \ (n \in \mathbb{N}_0),\ -\log p_n \sim cn^\beta\}.$$

Theorem 6.1 (Universality of each prescribed tail class)**.** *For every $\beta, c > 0$,*

$$\boxed{C_* = \sup_{P \in \mathcal{T}_{\beta,c}} \rho(P).} \tag{11}$$

The same C_* is the supremum over finitely supported nonnegative-integer laws with rational probabilities.

Proof. We give all approximation steps, since pointwise convergence alone would not justify entropy convergence.

Finite truncation. Given a finite-entropy law X, let F_m be an increasing finite exhaustion of its support and let X_m have the conditional law given $X \in F_m$. The formula

$$H(X_m) = \mathbb{P}(X \in F_m)^{-1} \sum_{x \in F_m} p_x \log(1/p_x) + \log \mathbb{P}(X \in F_m)$$

shows $H(X_m) \to H(X)$. Let $Z = (X, X')$, and let E_m be the event that both coordinates lie in F_m, of probability $1 - \delta_m \to 1$. Since $H(Z) < \infty$,

$$\delta_m H(Z \mid E_m^c) = \sum_{z \in E_m^c} \mathbb{P}(Z = z) \log \frac{1}{\mathbb{P}(Z = z)} + \delta_m \log \delta_m \longrightarrow 0.$$

For either $W = X + X'$ or $W = X - X'$, entropy decomposition gives

$$H(W) = (1 - \delta_m) H(W \mid E_m) + \delta_m H(W \mid E_m^c) + I(W; \mathbf{1}_{E_m}).$$

The second term is bounded by $\delta_m H(Z \mid E_m^c)$, and the last by the binary entropy $h(\delta_m)$. Thus $H(W \mid E_m) \to H(W)$; the conditional law is exactly $X_m \pm X_m'$. Positivity of Δ_+ gives convergence of the entropy ratios. This strengthens the elementary truncation principle used in [2, Lemma 7.5] to two-sided convergence for the present outputs.

Integer modelling and rational probabilities. For a finite real support A, take a basis of its rational linear span, express all its members in rational coordinates, and clear denominators. This identifies A additively with a finite subset of $\mathbb{Z}^r$. An integer linear form can be chosen injective on the union of this set, its sumset, and its difference set: finitely many nonzero difference vectors must have nonzero image, so avoid their finitely many linear hyperplanes. Explicitly, a form with coefficients $1, M, \ldots, M^{r-1}$ works for any integer M avoiding finitely many polynomial roots. Its image preserves all sum and difference collisions and hence all three entropies. Translation makes the support nonnegative. Approximate its positive probabilities by positive rational probabilities. Entropy is continuous on each relevant finite simplex, and $\Delta_+ > 0$, so the ratios converge.

Adding an arbitrarily prescribed tail. Let P be any finite nonnegative-integer law. Define a full-support law Q by

$$Q(n) = Z^{-1} \exp[-c(n+1)^\beta], \qquad n \geq 0.$$

Its entropy and all moments are finite. For $0 < \varepsilon < 1$ put $P_\varepsilon = (1 - \varepsilon)P + \varepsilon Q$. Beyond the finite support of P, its logarithmic tail is

$$-\log P_\varepsilon(n) = c(n+1)^\beta + \log Z - \log \varepsilon = cn^\beta + o(n^\beta).$$

Hence $P_\varepsilon \in \mathcal{T}_{\beta,c}$. For a finite mixture of finite-entropy laws with mixing weights w_i, its entropy lies between $\sum_i w_i H_i$ and $\sum_i w_i H_i + H(w)$. Apply this to the input mixture and the four-component mixtures describing its sum and difference outputs. All component entropies are finite, since $H(U \pm V) \le H(U) + H(V)$. It follows that all three entropies converge to those of P as $\varepsilon \downarrow 0$, and so do the ratios. Combining the three steps proves the theorem; the reverse supremum inequality is immediate. □

Corollary 6.2 (Same supremum, no Gaussian-tail maximizer)**.** *For every $c > 0$, the class $\mathcal{T}_{2,c}$ has supremum C_* but no member attaining it. More generally, an extremizer in any $\mathcal{T}_{\beta,c}$ can exist only when*

$$\log_2\left(\frac{4L}{1+L}\right) < \beta \le \log_2(8/3), \qquad L = 1.4051810329. \tag{12}$$

The endpoints are approximately 1.224610521648 *and* 1.415037499279*.*

Proof. The supremum equality is Theorem 6.1. An attainer would have $\rho = C_* > L > 1$, by (1). Finite-coordinate perturbations remain inside the same class, so Theorem 5.3 forces $\beta = \log_2(4C_*/(1+C_*))$. This expression is strictly increasing in $C_* > 0$. The known range (1) gives (12), which excludes $\beta = 2$. □

The supremum theorem does not force any particular tail shape: every prescribed tail can be added with arbitrarily small mass to an almost optimal finite law. The stationarity theorem, in contrast, restricts a single law that actually attains a local maximum. Keeping these two quantifiers distinct avoids an unjustified existence claim.

7 Verification record and remaining limitations

The package contains two standard-library verification programs:

```
python verify_midpoint.py
python verify_entropy_variation.py
```

The first checks all 19,683 labelled two-colour plane partitions, including the 1,371 admissible ones; 1,159 maximal-direction fibre and occupied-quotient instances; 640 plane-free projection instances; all 296,010 six-point subsets of $\mathbb{F}_3^3$ for the blocking claim; all 75,816 spanning five-point subsets for the sumset claim; and all 17,550 four-point subsets for the stated classification. It also verifies the sharp sixteen-point construction and the recurrence arithmetic. Some three-dimensional fibre checks are sampled products, not an exhaustion of all three-dimensional colourings. The exact five-point check finds a minimum sumset size of twelve; the written proof uses only eleven.

The entropy program checks 963 formal first-variation identities using exact rational coefficients of formal logarithms. It also certifies a finite fixed-support ascent: on $\{0, \dots, 8\}$, start with integer weights 2^{64-n^2} and transfer probability 10^{-12} from zero to six. The ratio increases from $1.2166978238640320\dots$ to $1.2166978239078707\dots$. This is a finite diagnostic, not a new bound for C_* or a certificate for an infinite Gaussian-tail law. Rational logarithm enclosures use the positive atanh series, ninety terms, its geometric tail bound, and outward rounding. The receipts retain the full weights, intervals, and positive gap. The infinite-support assertions rest on the proofs of dominated differentiation and convergence above, not numerical truncation.

The exploratory directory records a time-limited four-dimensional mixed-integer search that found no feasible point. That outcome is neither an infeasibility proof nor evidence of an exact extremal value.

The general $O(k \log k)$ midpoint codimension bound and the entropy range (1) remain unchanged. The new midpoint theorem does not settle higher-flat-free remainders or the exact two-colour

extremal function. The tail condition is necessary, not sufficient, and does not establish an entropy maximizer or rule out irregular or sparse extremizers.

A Proof of the prior cap classification

A plane cap has at most four points. Through a fixed cap point there are four lines, each containing at most one further cap point, so initially the bound is five. Equality would make every line through any cap point meet the cap twice; no line could meet it once. But three parallel lines would partition five points into sizes $2, 2, 1$, a contradiction. For a four-point cap, put three points at $0, e_1, e_2$. The fourth can only be $e_1 + e_2$, $e_1 - e_2$, or $-e_1 + e_2$, each giving an affine parallelogram.

No plane section of an admissible two-colour cap remainder can have exactly three points. Put such points at $0, e_1, e_2$. The coloured points $e_1 + e_2, e_1 - e_2, -e_1 + e_2$ have these three points as their pairwise midpoints, forcing three colours.

Now let W be an affine three-space, $s = |R \cap W|$, and let n_j count its planes with j remainder points. Possible section sizes are $0, 1, 2, 4$. The 39 planes, 13 planes through each point, four planes through a pair, and one plane through a noncollinear triple give

$$n_0 + n_1 + n_2 + n_4 = 39, \qquad n_1 + 2n_2 + 4n_4 = 13s,$$
$$n_2 + 6n_4 = 4\binom{s}{2}, \qquad 4n_4 = \binom{s}{3}.$$

Three parallel planes show $s \leq 12$. Solving gives

$$n_4 = \tfrac{1}{4}\binom{s}{3}, \quad n_1 = 13s - 8\binom{s}{2} + 2\binom{s}{3}, \quad n_0 = 39 - 13s + 4\binom{s}{2} - \tfrac{3}{4}\binom{s}{3}.$$

For $s = 3, 5, 7, 11$, n_4 is not an integer. For $s = 6, 8, 9$, the value of n_1 is respectively $-2, -8, -3$. For $s = 10, 12$, the value of n_0 is respectively $-1, -18$. The only possibilities are $s = 0, 1, 2, 4$, with $n_4 = 1$ in the last case. Thus any four remainder points in a three-space are coplanar. Four affinely independent remainder points in the original ambient space would contradict this conclusion in their affine span. Therefore the whole remainder has affine dimension at most two, and the plane classification proves Lemma 2.1. This rederives the necessary classification from [6]; no new priority is claimed for this appendix.

References

[1] T. Kościuszko, *Arithmetic Progressions in Midpoint Colourings*, arXiv:2610.11847v1 (2026), Problem 4 and Theorem 3. `https://arxiv.org/html/2610.11847v1`.

[2] R. Li, *The Entropic Sum-Product Phenomenon*, arXiv:2607.29042v1 (2026), Lemma 7.5 and Section 8. `https://arxiv.org/html/2607.29042v1`.

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[5] B2 research continuation I, *Research note*, 13 October 2026. Supplied artifact `B2-new-results-2026-10-13.zip`.

[6] B2 research continuation II, *Two-colour rigidity and boundary obstructions for entropy extremizers*, 13 October 2026, Theorem 2.1, Lemma 3.1, and exact entropy certificate. Supplied artifact `B2-further-results-2026-10-13.zip`.